

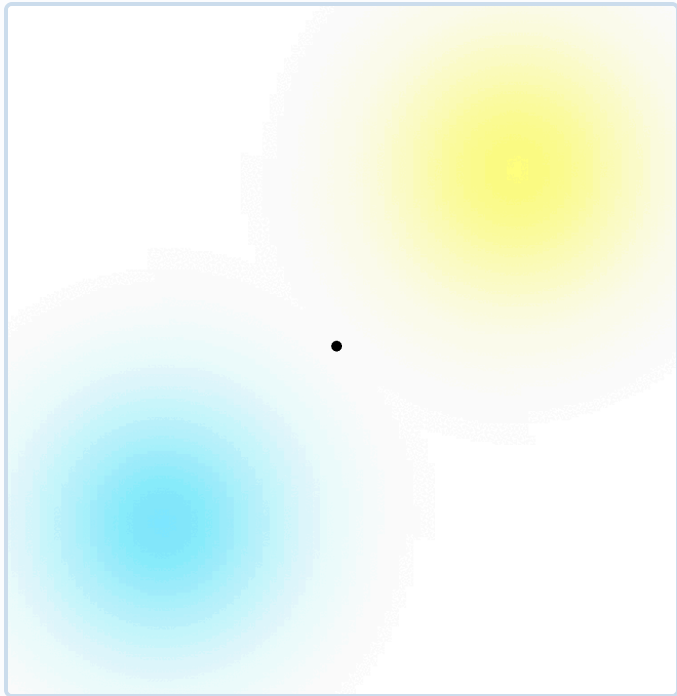
An operating system for the collective

Paper — *Towards Collective Robotic Operating Systems through Aggregate Computing*

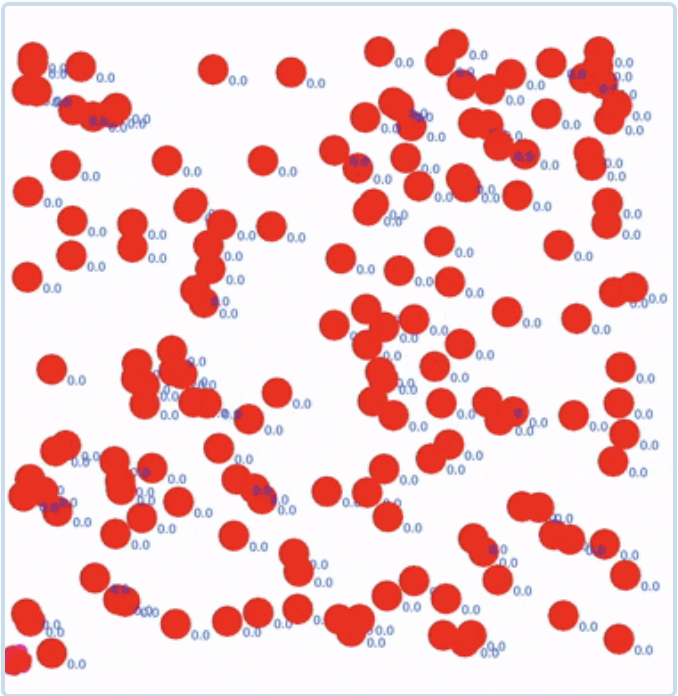
One program for the whole swarm — running on devices that move, fail, join and leave. **Aggregate Computing** as the runtime layer.

COLLECTIVE APPLICATIONS specified at the swarm level

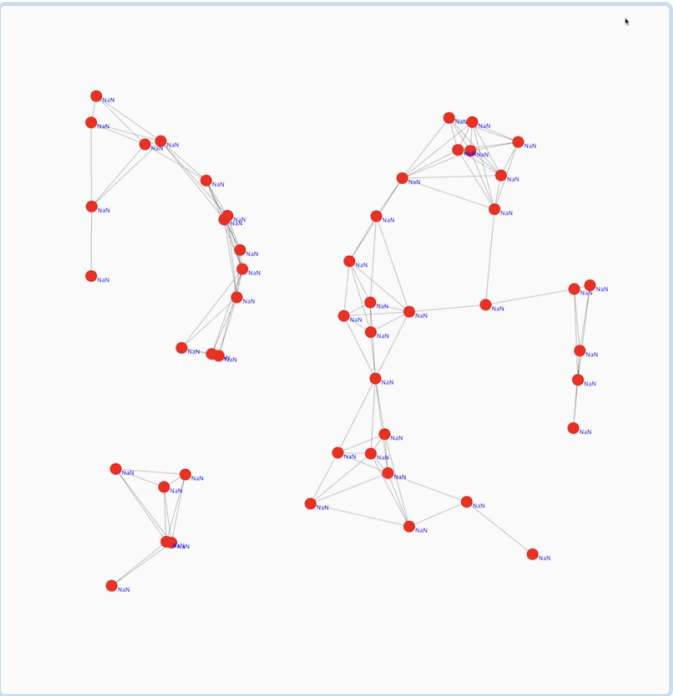
SELF-ORGANISING SERVICES — designed, implemented in Collekative, published



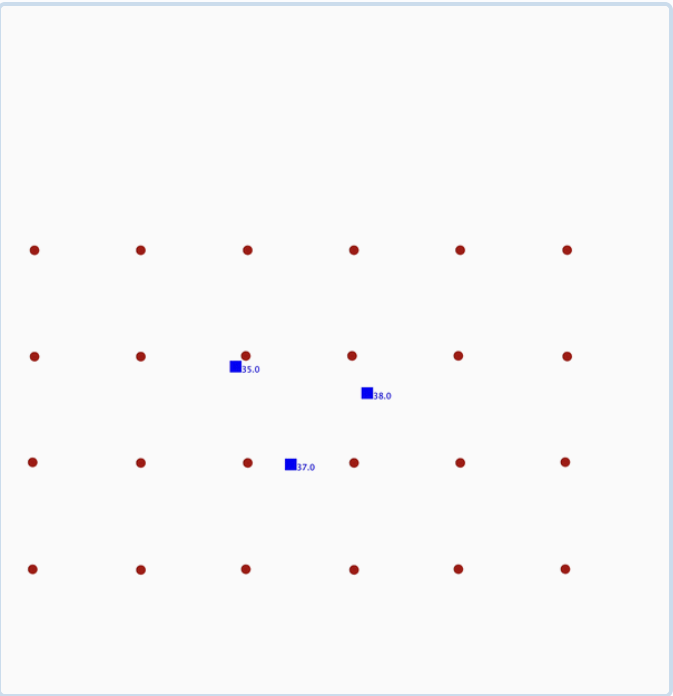
Formation
self-organising morphogenesis



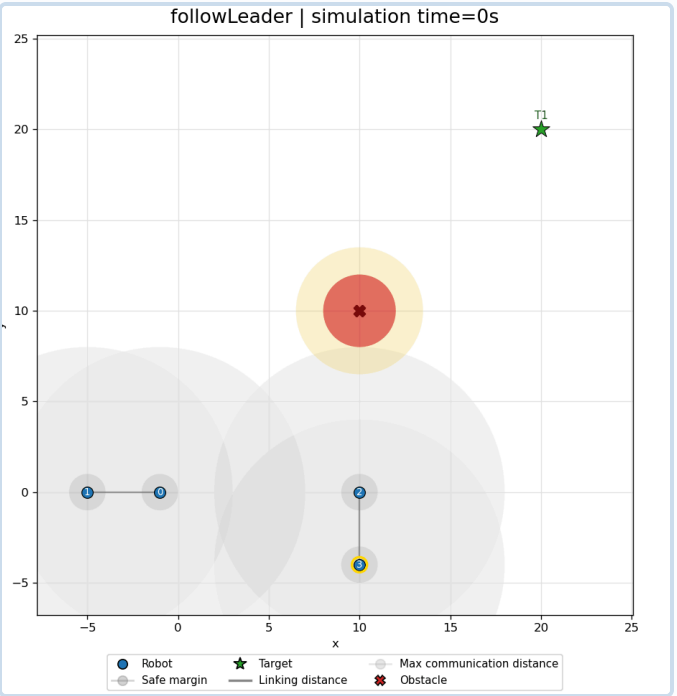
Replanning
tasks redistributed on failure



Consensus
self-stabilising min-max gossip



Sensing
distributed particle filtering



Safety
CBF/CLF filter on transients

DEVICES: MOBILE, HETEROGENEOUS, FAILING no central point of coordination